

# Five Languages, Five Bottlenecks: Diacritization and G2P for Transliteration at Interscript

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## Abstract

We modernized the phonological layer of the Interscript transliteration platform across five languages: Arabic, Hebrew, Thai, Persian, and Urdu, covering both diacritization (vowel restoration in the original script) and G2P (romanization/phonemization). Our central finding is a *bottleneck taxonomy*: each language’s performance ceiling was set by exactly one resource—data *quality* (Arabic), data *domain* (Hebrew), labeled-data *coverage* (Thai), output *representation* (Persian, Urdu)—and a fixed ByT5-based framework with one surgical fix per language reached or beat the published state of the art in each. Results: Arabic **2.81% DER** (CE) / 1.69% (w/o CE) on the SadeedDiac-25 benchmark with Misraj’s own evaluator under a zero-skip windowed protocol— $2.6\times/3.1\times$  better than the 1.5B-parameter Sadeed model, ahead of GPT-4 and Gemini-Flash-2.0 (only Claude-3.7 ahead), plus a contamination audit showing the benchmark’s public source corpus leaks 122 of its own paragraphs; Hebrew 17.5% DER on Biblical text where the SOTA model scores 35.6%; Thai 2.32% PER via deterministic (non-LLM) augmentation; Persian homograph accuracy 77.3–89.5% and diacritization 0.52% CER from a G2P-only dataset; Urdu G2P 14.77% CER (first learned baseline,  $4.1\times$  over rule-based) and diacritization 3.74% CER from weak supervision. We distill cross-cutting lessons for phonological NLP: LLMs are unreliable phonological labelers; evaluation domains and input formats silently dominate comparisons; ensembling and curricula fail at data-limited optima; and byte-level models are the right deployment target for multilingual TS runtimes.

## 1 Why phonology for transliteration?

Transliteration maps (ALA-LC, SBL, RTGS, ...) assume *vocalized* input. Undiacritized Arabic or Hebrew, or unsegmented Thai, cannot be transliterated unambiguously. The models described here supply the vocalization that makes interscript.org’s maps deterministic.

## 2 System and results

Common framework: ByT5 seq2seq (byte-level; tokenizer = UTF-8 bytes, so TS inference needs no vocabulary assets) except Arabic, where a compact 30M char-level encoder won on cost.

## 3 The bottleneck taxonomy

- **Arabic — data quality:** cleaning +  $28\times$  scaling halved DER; a 30M encoder matches a 1.5B LM.
- **Hebrew — data domain:** the SOTA model collapses cross-domain ( $9\times$ ); mixed-domain training wins; the teamim input-format effect makes cantillation copy-through.

| Language | Task           | Result                       | Published reference                                              |
|----------|----------------|------------------------------|------------------------------------------------------------------|
| Arabic   | diacritization | <b>2.81/1.69% DER</b>        | Sadeed 7.29/5.26%, Gemini 3.19/2.38% (same benchmark+evaluation) |
| Hebrew   | diacritization | <b>17.46% DER</b> (Biblical) | DictaBERT 35.6% (same test)                                      |
| Thai     | G2P            | <b>2.32% PER</b>             | baseline 6.37% (same test)                                       |
| Persian  | G2P/HA         | <b>89.5% / 77.3% (SB)</b>    | Homo-GE2PE 76.9%                                                 |
| Persian  | diacritization | <b>0.52% CER</b>             | (none existed)                                                   |
| Urdu     | G2P            | <b>14.77% CER</b>            | epitran 60.0%                                                    |
| Urdu     | diacritization | <b>3.74% CER</b>             | (none existed)                                                   |

- **Thai — labeled coverage:** 10K dictionary  $\rightarrow$  60K via deterministic phonemizer labels; all architectural fixes failed, data fixed it.
- **Persian — representation:** one dataset serves two tasks (G2P + diacritization); metric representation choices (ezafe) shift results by 18 points.
- **Urdu — supervision existence:** no gold diacritization exists; weak IPA-derived labels at 597K scale suffice.

## 4 Cross-cutting lessons

1. **LLMs cannot label phonology.** Tone and haraqat hallucination mirrors the known diacritization failure; epitran labels are free of it and transfer through fine-tuning.
2. **Input formats and domains are hidden variables.** Teamim-preserving inputs and Biblical-vs-modern test domain each move DER by more than most modeling choices.
3. **Beam search is a first-class factor** (12 DER points on Hebrew); ensembles must be evaluated under beam search to be meaningful.
4. **Output-vote ensembling hurts seq2seq restoration:** per-character voting lowers error counts yet raises edit-distance metrics (incoherent splices).
5. **Curricula fail at data-limited optima;** their reported gains live in pretraining regimes.
6. **Byte-level models are deployment-friendly:** ByT5’s tokenizer is `TextEncoder` in TypeScript—no vocab files, no WASM sentencepiece—ideal for interscript.org’s browser/Node targets.

## 5 Deployment

Best checkpoints export to ONNX; distribution follows the Interscript plan (GitHub Releases primary, HuggingFace canonical, jsDelivr edge). The TS package wraps `onnxruntime-web/node` with byte-level tokenization.

## 6 Reproducibility

Per-language repositories (`rababa`, `rababa-farsi`, `rababa-urdu`, `secryst`) contain training pipelines, evaluation harnesses, `docs/RESULTS.md` ground truth, and the five companion papers in `docs/paper-*/`.